

Case Study 16

(a) identify and justify the nature of the dataset

The given dataset represents the Crop yield (tonnes/hectare) recorded for five districts across four quarters (Q1-Q4). Hence it should be treated as a two-dimensional (matrix-structured) dataset.

The two dimensions are:

- Rows: District (Thanjavur, Cuddalore, Erode, Salem, Coimbatore)

- Columns: Quarters (Q1, Q2, Q3, Q4)

Each cell in the matrix represents the Crop yield for one district during one quarter.

This classification is important because different data structures require different visualization techniques. Since the dataset contains two categorical variables (district and quarter) and one numerical value (Crop yield), a heatmap is the most suitable visualization.

A heatmap allows quick comparison across both dimensions and makes it easier to identify high yield and low-yield areas using colour intensity.

If the dataset were treated as a simple one-dimensional dataset,

valuable relationship between districts and seasons would be lost. Therefore, recognizing it as a matrix dataset helps in selecting the correct visualization and enables better analysis and decision-making.

1) Comparison heatmap and a grouped bar chart

Both a heatmap and a grouped bar chart can be used to represent

this dataset, but each has its own advantages and disadvantages.

Heatmap

Strengths

It provides an immediate visual comparison across all districts and quarters using

Colour intensity

Weaknesses

- 1) Exact numerical values cannot always be read accurately without displaying annotations
- 2) People with color vision deficiencies may find it difficult to distinguish certain

Colour shades

Grouped Bar chart

Strengths

Exact group yield values can be compared easily using the heights of the bars.

- 2) It is familiar to most users and easy to understand

Weaknesses

- 1) With too many districts and four quarters, the chart becomes crowded with many data

Since the dataset contains a mean of district and quarter, a heatmap is more suitable.

Subplots become a high overall pattern and comparison.

- c) Analyse the dataset and describe key patterns

The dataset shows the difference in crop yield across five districts and four quarters.

Highest crop yield: 6.6 tonnes/hectare, recorded in Salem (Q₁) and Thanjavur (Q₄)

Lowest crop yield: 2.6 tonnes/hectare, recorded in Salem (Q₃)

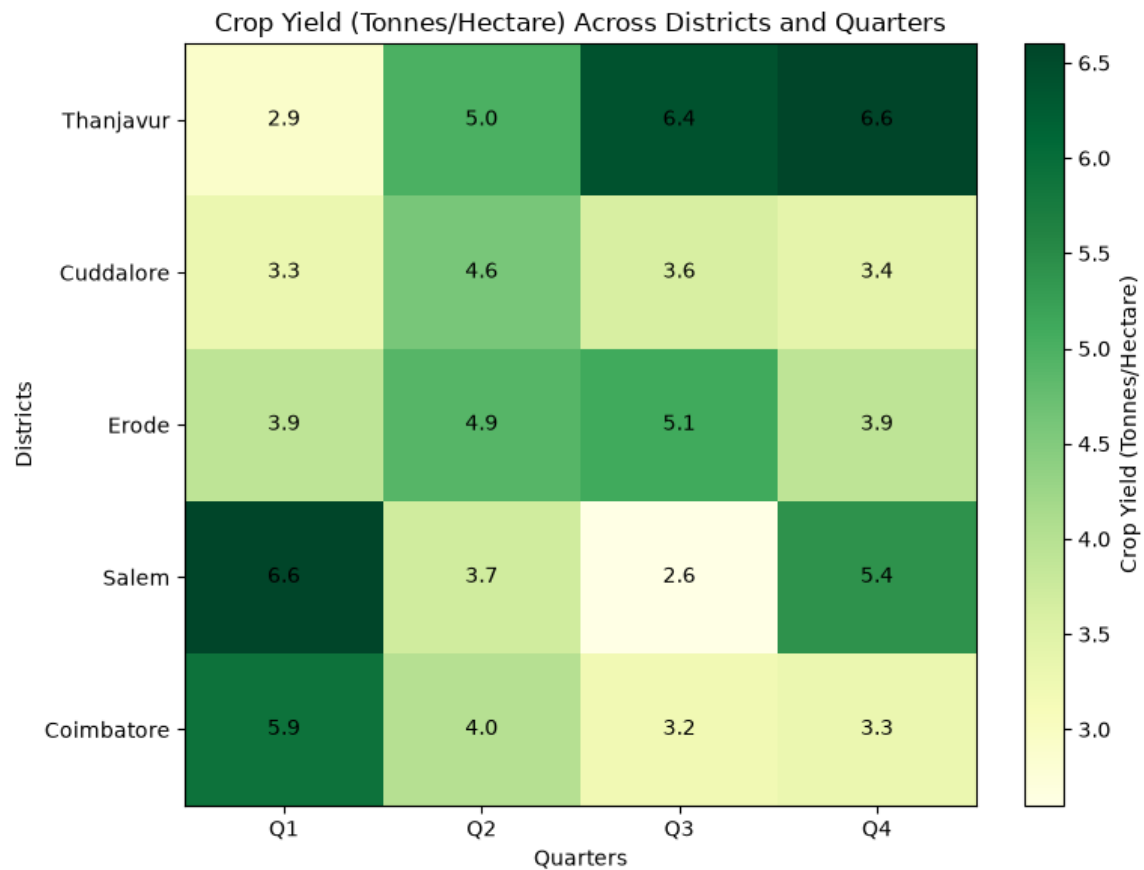
Key observations

- Thanjavur shows a steady increase from 2.9 in Q₁ to 6.6 in Q₄
- Cuddalore remains fairly consistent, with yields between 3.7 and 4.8 tonnes/hectare
- Erode reaches its highest yield (5.1) in Q₃
- Salem has the greatest variation, dropping from 6.6 in Q₁ to 2.6 in Q₃, before recovering to 5.4 in Q₄
- Coimbatore shows a gradual decline from 5.9 in Q₁ to 3.2 in Q₄

Conclusion

The dataset indicates clear seasonal variation. Thanjavur shows continuous improvement, Cuddalore is stable, Salem has the most fluctuation, and Coimbatore experiences a gradual decline. These observations can help the Co-operative Society plan future

farming strategies and resource allocation.



Heatmap is recommended because:

1. It clearly shows the variation in crop yield across both **districts** and **quarters** using color intensity.
2. The **highest yields (6.6)** and **lowest yield (2.6)** are immediately identifiable.
3. It allows quick comparison of seasonal patterns across all districts without clutter, making it suitable for management decision-making.

Explain how you would improve the visualization.

To make the visualization more useful for the Cooperative Society president, the

following improvements can be made.

Use an appropriate colour scale

Apply a light-to-dark colour gradient where darker colours represent higher

crop yields and lighter colours represent lower yields

Display numerical values

Show the exact crop yield inside each heatmap cell so user can read

Precise values without estimating

Add a colour legend

include a legend indicating the crop yield corresponding to each colour

improves layout

Clearly label the district on the vertical axis and quarter on the horizontal axis

with readable font size

Highlight important values

use bold borders and different colours to emphasize the highest and lowest

crop yields

Provide interactive features

Allow users to hover over cells to view detailed information such as

district names, quarter, and crop yield

a) Identify and justify the nature of the dataset

The given dataset contains independent samples of crop yield (tonnes/hectare) collected from three different districts: Thangjam, Cuddalore and Erode. Therefore, it should be treated as a multi-group distribution dataset.

Each district represents an independent group and multiple group yield observation have been recorded for each group. The objective is to compare the distribution of crop yields across these districts rather than a single value.

This classification is important because the data consists of several groups with continuous numerical values. A violin plot or boxplot is more appropriate than a bar chart since these charts display the distribution.

b) Compare a violin plot and a boxplot

Both the violin plot and the boxplot can be used to compare group yield distribution across the three districts.

Violin plot

Strengths

1. It displays the full distribution and density of the crop yield values for each district.
2. It clearly shows whether values are concentrated or spread across different yield ranges.

Weakness

It is more difficult to interpret for users unfamiliar with violin plots. Exact quartile values are less obvious than in a boxplot.

Box plot

Strengths

1. It clearly displays the median, quartiles and overall spread of data.
2. It easily identifies possible outliers.

Weakness

It does not show the shape or density of the distribution. Different distributions may appear similar if they have similar quartiles.

c) Analyse the dataset

The dataset shows the variation in crop yield among the three districts.

* Highest Crop yield 6.1 tonnes/hectare, recorded in all three districts.

* Lowest Crop yield 2.4 tonnes/hectare, recorded in Thanjavur.

Key observation

* Thanjavur has the widest spread ranging from 2.4 to 6.1, indicating high variability.

* Cuddalore shows a gradual increase with several observations at 6.1, indicating consistently high values, and Erode shows stable crop yields with moderate variation.

e) State one limitation and explain how could improve it

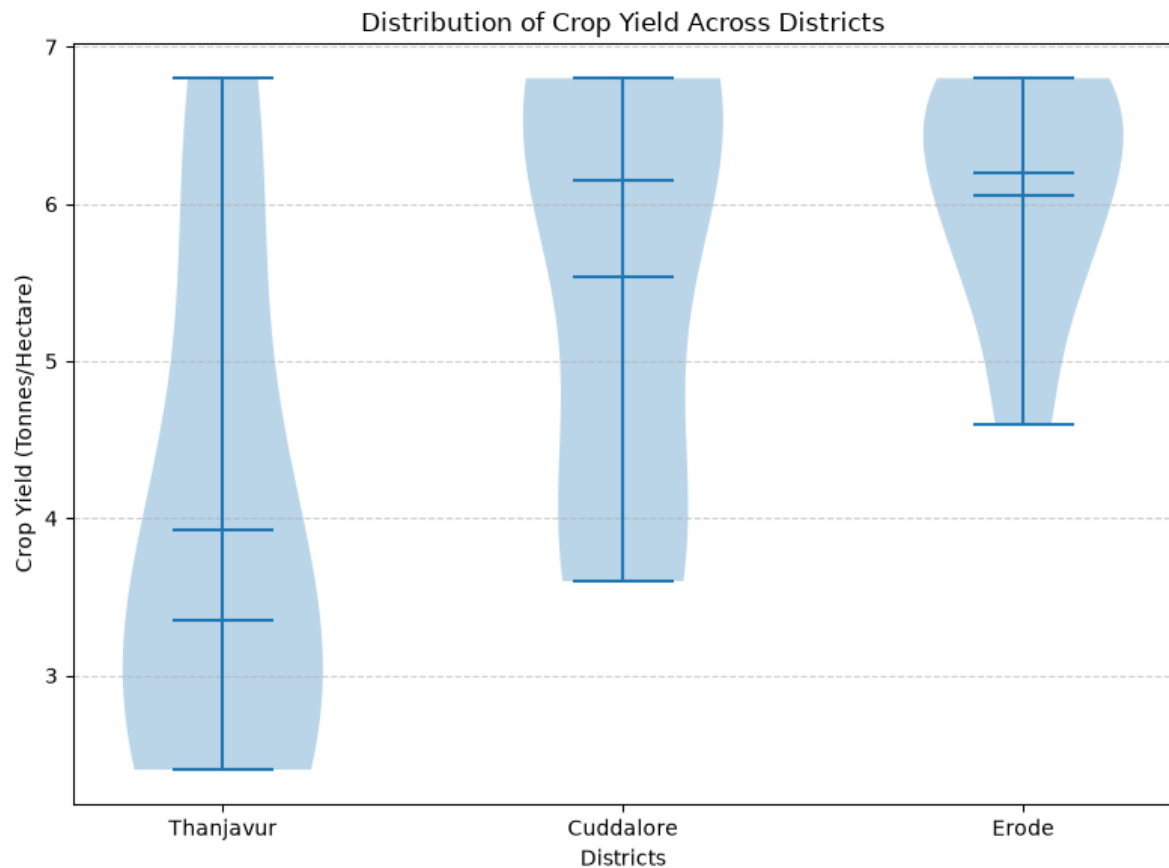
One limitation of the violin plot is that it may be difficult for some users to interpret because it emphasizes the distribution shape rather than exact numerical values.

This limitation can be addressed by:

1) Adding the median and quartile lines inside the violin plot

2) Display the exact crop yield values as data labels or annotations

3) Combining the violin plot with a boxplot to show both the distribution and summary statistics.



Justification (Write below the visualization)

1. A violin plot shows the full distribution of crop yields for each district, not just summary statistics.
2. It clearly highlights differences in the spread and density of values among Thanjavur, Cuddalore, and Erode.
3. It makes it easier to compare the variability and concentration of crop yields across districts than a simple boxplot, making it more suitable for this distribution-focused analysis.